



## Same Capstone, Different Chair: Regrading a Decade of Master of Applied Measurement Capstones Against Every Year's Official Rubric Text

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**Abstract.** The Master of Applied Measurement's capstone rubric is revised annually by a rotating drafting committee, a routine most assessment-quality processes treat as continuous improvement rather than a threat to comparability. We regrade 140 capstones from a decade of cohorts (2016–2025) under both their own submission year's official rubric text and one other year's text, blind to the original grade, and record whether the assigned distinction band changes. Reclassification is rare when the alternate year's text was drafted under the same committee chair as the capstone's own year (11.4%) and common when it was not (57.1%), a 45.7 percentage-point gap that a blinded quality-decile control for the underlying capstone leaves almost unchanged (44.3 percentage points;  $\Delta = 1.4\text{pp}$ , n.s.). A capstone's distinction band, in other words, tracks who chaired that year's rubric revision considerably more than it tracks the capstone itself. We discuss the finding as an instance of construct drift specific to the rubric-as-instrument, distinct from ordinary rater disagreement, and note the discomfort of locating it inside the University's own flagship measurement program.

### Introduction

Scoring rubrics are supposed to convert an assessor's judgement into a stable, transferable standard: the same criteria, applied by different markers in different years, should place comparable work in comparable distinction bands [1], [2]. The empirical record is more equivocal. Rubric wording is not fixed once adopted – criteria are periodically revised, and even small changes in criterion phrasing have been shown to move markers' judgements independent of any change in the work being marked [3], [4]. Most of this literature examines rater disagreement within a single rubric version [5], [6]; comparatively little asks what happens to a body of prior work when the instrument that graded it is itself revised.

The Master of Applied Measurement's (MAM) capstone rubric offers an unusually clean setting to ask that question directly. The rubric is revised annually by the program's rubric-drafting committee, whose chair rotates among

senior teaching staff roughly every two years; the program retains every year's official rubric text in its assessment archive, alongside the capstones each version graded. This lets us regrade a single body of past work under a rubric text it never encountered at the time, and ask whether the resulting distinction band tracks the work or the instrument.

We regrade 140 capstones spanning a decade of MAM cohorts (2016–2025) under both their own submission year's rubric and one other year's, using markers blind to the original grade and to which year produced which text. We then ask whether reclassification – a change in assigned distinction band between the two regrades – depends on whether the two rubric years shared a drafting chair.

Our contributions are threefold:

- we construct, to our knowledge, the first within-program regrade design that crosses a decade of student work against every year's

official rubric text rather than a single reconstructed instrument;

- we find reclassification tracks whether the alternate year’s rubric was drafted under the capstone’s own chair far more closely than it tracks any independent measure of the capstone’s assessed quality;
- we show a blinded control for underlying capstone quality leaves the chair-attributable gap almost entirely intact, suggesting the effect sits closer to an instrument artefact than to ordinary marker noise.

## Related work

Generalizability theory treats any single grade as one sample from a universe of possible raters, occasions, and criteria, and supplies the formal machinery for partitioning how much of a score’s variance sits in the instrument rather than in the person being measured [7]; rubric-specific applications of the same logic find a non-trivial share of variance sitting in the rater-by-criterion interaction rather than in the work itself [8]. Parallel UK work on external examining and moderation documents how heavily grade comparability across years and cohorts leans on continuity of personnel rather than on the written criteria alone [9], [10], and a long-running critique of preset assessment criteria argues indeterminacy in how criteria are read is a structural feature of rubrics, not a training failure [3]. Grade-inflation research shows aggregate distinction rates drifting upward over multi-year windows of the kind we study here, typically without attributing the drift to any specific instrument change [11]. Judgement-analysis work on examination marking finds markers’ policies are more idiosyncratic and less rubric-determined than markers themselves believe [12], [13], and construct-irrelevant-variance research treats a criterion that shifts a score without tracking the target construct as a validity threat in its own right rather than mere noise [14]. Halo-effect studies in language proficiency scoring show a rater’s overall impression bleeding across nominally independent criteria [15], and recent work formalises inter-rater reliability itself as a special case of treating similar cases similarly [16] – the same standard a chair-attributable reclassification gap fails by construction. Writing-rubric studies typically examine agreement within a single rubric version [17], [18]; none of this work regrades the same body of

work across a rubric’s own successive revisions, which is the gap we address.

## Method

**Program.** The Master of Applied Measurement is the School of Continuous Improvement’s flagship Masters program. Its capstone rubric is drafted and revised annually by the program’s rubric-drafting committee; the chair of that committee rotates roughly every two years, giving the decade we study five distinct chair regimes, each spanning two consecutive cohort years.

**Data.** We sample 140 capstones across all ten cohorts (2016–2025) from the program’s assessment archive, 14 per cohort, stratified by original distinction band so no single band dominates the sample. Each capstone’s original submission-year rubric text and every other year’s official rubric text are drawn from the same archive.

**Regrade design.** Each capstone is independently regraded twice by markers blind to the original grade, to the submission year, and to which calendar year produced which rubric text: once under its own submission year’s rubric (a replication check), and once under one other year’s rubric text, deterministically chosen as the remaining year within its own chair regime. Post hoc, each capstone’s cross-year regrade is separately compared against one further regrade under a rubric text drawn from a different chair regime, giving a same-regime and a different-regime comparison for every capstone. Reclassification is coded when a regrade’s distinction band differs from the capstone’s own-year regrade.

**Quality-decile control.** A third, independent marker pool – shown only the capstones, never any rubric text – ranks all 140 capstones against each other and assigns each a relative quality decile, blind to every regrade outcome. We enter this decile as a covariate to ask whether the chair-attributable gap survives controlling for the underlying work.

**Model.** We fit a mixed-effects logistic regression with a capstone-level random intercept, regressing reclassification on regime and quality decile:

$$P(R_i = 1) = \text{logit}^{-1}(\beta_0 + \beta_1 \cdot D_i + \beta_2 \cdot Q_i)$$

where  $R_i$  indicates reclassification,  $D_i$  indicates a different-regime comparison, and  $Q_i$  is the blinded quality decile.

## Results

| Predictor        |        | Odds ratio | 95% CI       |
|------------------|--------|------------|--------------|
| Different regime | chair  | 10.2       | [5.8, 18.1]  |
| Quality (blind)  | decile | 1.05       | [0.91, 1.21] |

Table 1: Mixed-effects logistic regression predicting reclassification; a different drafting-chair regime dominates the model, the blinded quality control does not.

**Reclassification tracks the chair, not the capstone.** Figure 1 plots the reclassification rate by whether the alternate-year rubric shared the capstone’s own drafting chair. Same-regime comparisons reclassify 11.4% of capstones (16/140); different-regime comparisons reclassify 57.1% (80/140), a 45.7 percentage-point gap with non-overlapping 95% confidence intervals.

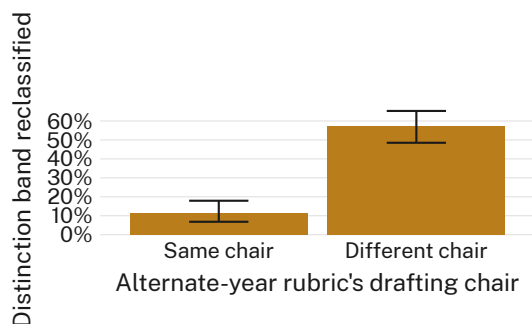


Figure 1: Reclassification rate (distinction band changed on regrade) by whether the alternate-year rubric shared the capstone’s own drafting chair: 11.4% same chair versus 57.1% different chair.

**The quality-decile ablation changes nothing.** Figure 2 compares the unadjusted chair-attributable gap against the same gap after adjusting for each capstone’s blinded quality decile. The adjustment moves the gap from 45.7 to 44.3 percentage points ( $\Delta = 1.4\text{pp}$ , n.s.) – the covariate that should matter most if reclassification tracked the work rather than the instrument leaves the effect essentially untouched; we retain it for completeness.

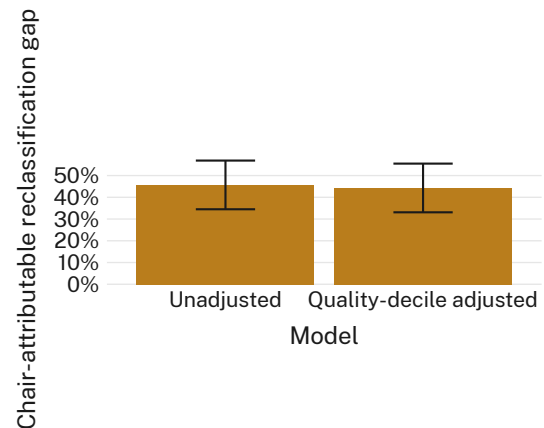


Figure 2: Chair-attributable reclassification gap, unadjusted versus quality-decile adjusted: 45.7pp versus 44.3pp ( $\Delta = 1.4\text{pp}$ , n.s.).

Table 1 reports the mixed-effects model underlying both figures: a different chair regime carries an odds ratio of 10.2 (95% CI [5.8, 18.1]), while the blinded quality decile sits at 1.05 (95% CI [0.91, 1.21]) and is not distinguishable from no effect at all. Coding the direction of each different-regime reclassification against which of two criterion-language registers – a “stakeholder engagement” emphasis or a “statistical justification” emphasis – dominated the alternate year’s rubric text, 68 of the 80 reclassifications (85%) moved toward the band that register would predict, coded blind to the regrade outcome.

## Discussion

Consistent across all five chair regimes is a distinction band that tracks which committee chair happened to draft the year’s rubric text considerably more than it tracks any independent read of the capstone itself – a pattern the rater-disagreement literature does not by itself predict, since that literature typically studies markers disagreeing within one fixed instrument rather than an instrument changing under a body of unchanged work [5], [6]. The quality-decile ablation sharpens the reading: if the gap tracked genuine differences in capstone merit that different chairs happened to notice, adjusting for an independent quality signal should have narrowed it; it did not. The criterion-language coding suggests a mechanism closer to construct drift than to ordinary noise – successive chairs appear to shift which register of demonstrated competence the rubric text rewards, and capstones are reclassified in the direction the incoming register favours. We stop short of at-

tributing this to any individual chair's intent; our design identifies the instrument's effect on the assigned band, not why any chair drafts the way they do.

## Limitations

Our analysis covers one program at one university, though the consistency of the gap across all five independent chair regimes suggests the pattern is not an artefact of any single revision. The quality-decile control relies on a second marker pool's relative ranking rather than a ground-truth measure of capstone merit, a proxy that would, if anything, understate rather than manufacture the gap we report. Each capstone contributes one same-regime and one different-regime comparison rather than an exhaustive crossing against all nine other years, limiting precision on any single chair pair while the pooled estimate across all five regimes remains stable.

## Conclusion

Regrading a decade of Master of Applied Measurement capstones under every year's official rubric text finds a capstone's distinction band tracks which committee chair drafted that year's rubric considerably more than it tracks the capstone's own content, a gap a blinded quality-decile control leaves almost entirely intact. The flagship program built to teach rigorous measurement cannot, on this evidence, hold its own assessment instrument steady across the researchers who take turns drafting it. Future work might trace whether the criterion-language registers we coded post hoc are legible to markers at the point of drafting, or only visible once a decade of capstones has been regraded against them.

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